

Technology Trends: Current VS Future

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OUTLINE



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EXECUTIVE SUMMARY



- Demographic Gaps in Developers:
 - Gender
 - Location
 - Age
- Trends in Technology:
 - Current popular languages, databases, platforms and frameworks used by developers.
 - Desired languages, databases, platforms and frameworks for developers.

INTRODUCTION



- This is aimed to analyze the current and desired technologies and their demographic analysis.
- The analysis should aid for decision-making on which technologies to invest in regarding to train current developers.
- Also, the location of most developers along with it demographic information is highlighted to aid for hiring or training.



METHODOLOGY



- Datasets:
 - Programming Languages and Annual Salary.
 - Github Jobs API.
 - Stack Overflow Developer Survey.
- Data Collection Methods:
 - APIs and Web Scrapping.
- Data Cleaning:
 - Removing duplicates.
 - Inputting missing values.
 - Normalizing data.
- Data Exploration and Visualization:
 - Python (Pandas, Matplotlib & Seaborn).
- Dashboard:
 - IBM Cognos Analytics.



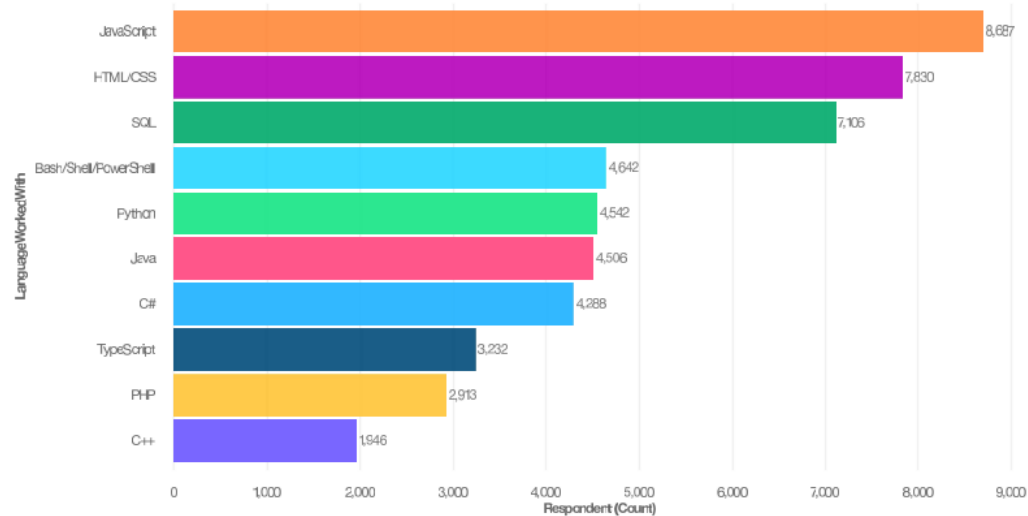


RESULTS

PROGRAMMING LANGUAGE TRENDS

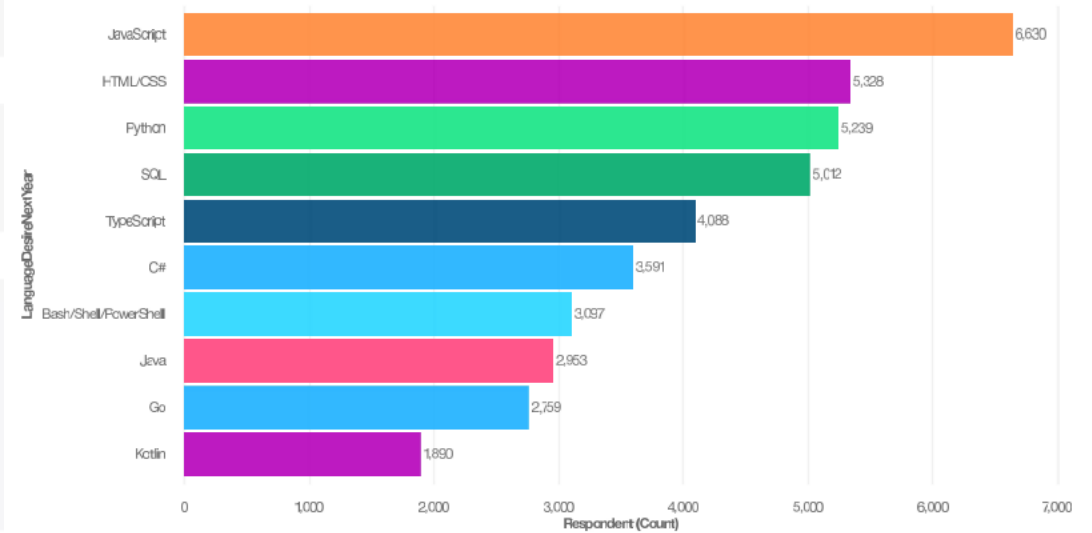
Current Year

Top 10 Languages Worked With



Next Year

Top 10 Languages Desired to Work With Next Year



PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS

Findings

- Javascript and HTML/CSS are the two most popular languages.
- Python popularity is increasing indicating that it moves towards Data Analysis, Data Science, AI and Machine Learning (ML).
- Kotlin, a mobile development language, is increasing its interest.

Implications

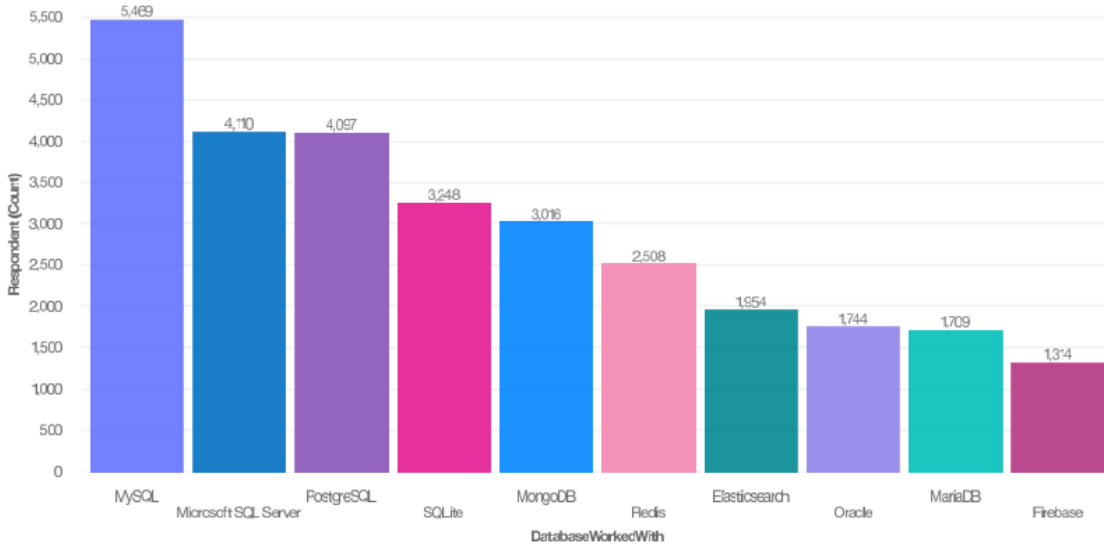
- Investing for Data Analysis, Data Science, Machine Learning and AI.
- Continue in training for Javascript and HTML/CSS for developers.

DATABASE TRENDS

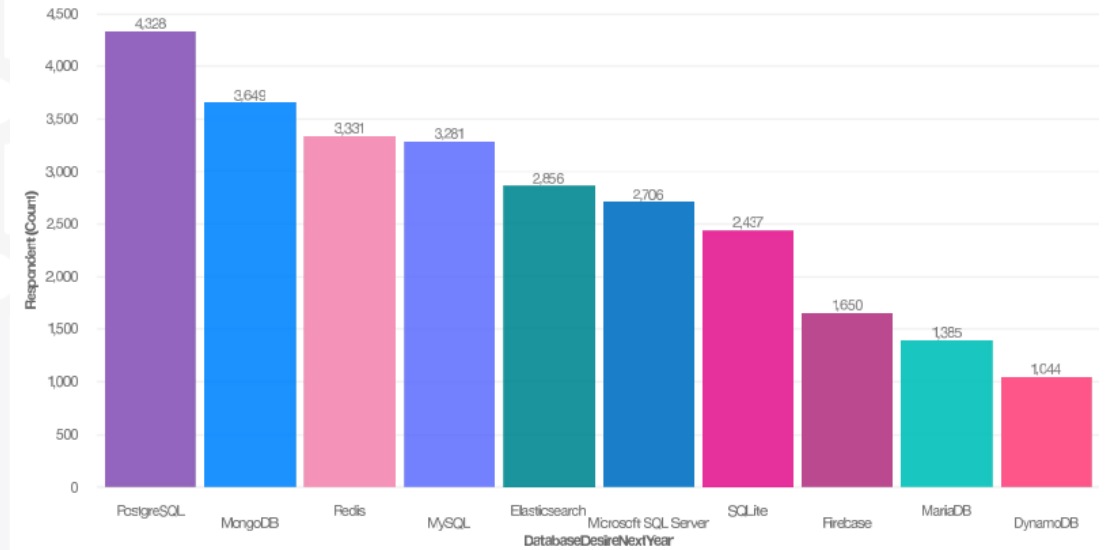
Current Year

Next Year

Top 10 Databases Worked With



Top 10 Databases Desired to Work With Next Year



DATABASE TRENDS - FINDINGS & IMPLICATIONS

Findings

- RDBMS are the most popular databases.
- MongoDB is gaining popularity for Non-Relation DBs.
- Oracle databases are not popular nowadays.

Implications

- Invest in a modern DBMS like PostgreSQL & MongoDB.
- Stop training or invest less in training for developers experienced in Oracle.



DASHBOARD

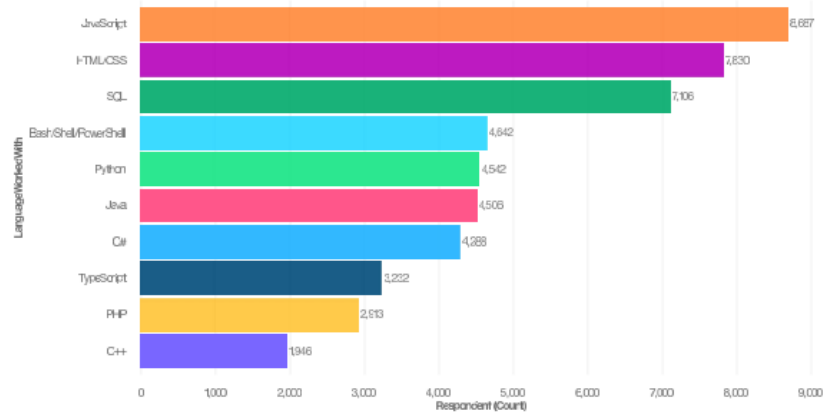


https://github.com/betofrr/IBM-Data-Analyst-Capstone-Project/blob/main/IBM_Cognos_Dashboards.pdf

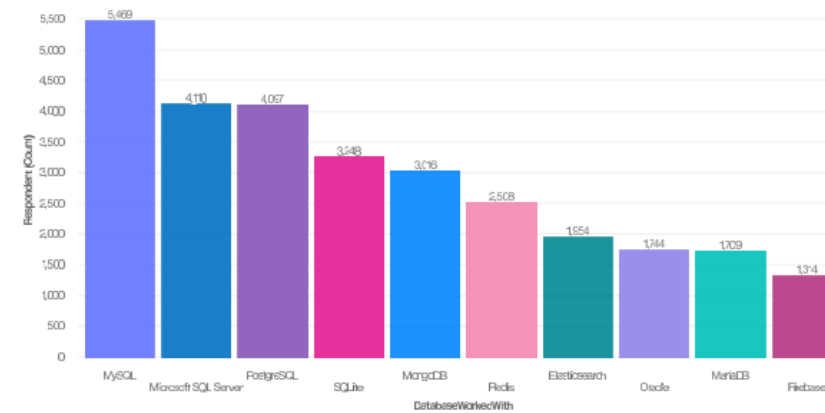


DASHBOARD TAB 1

Top 10 Languages Worked With



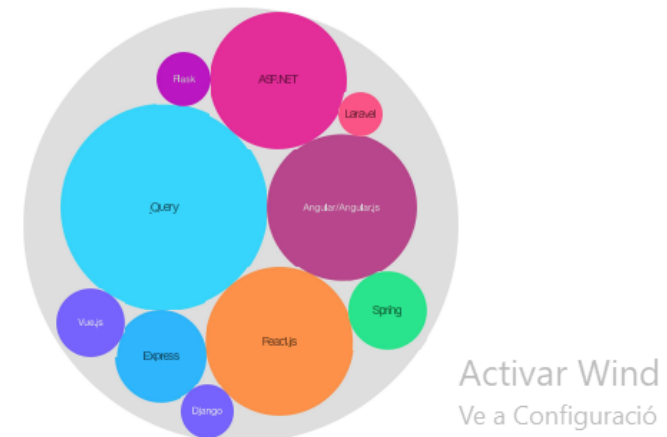
Top 10 Databases Worked With



Platforms Worked With



Top 10 Webframes Worked With

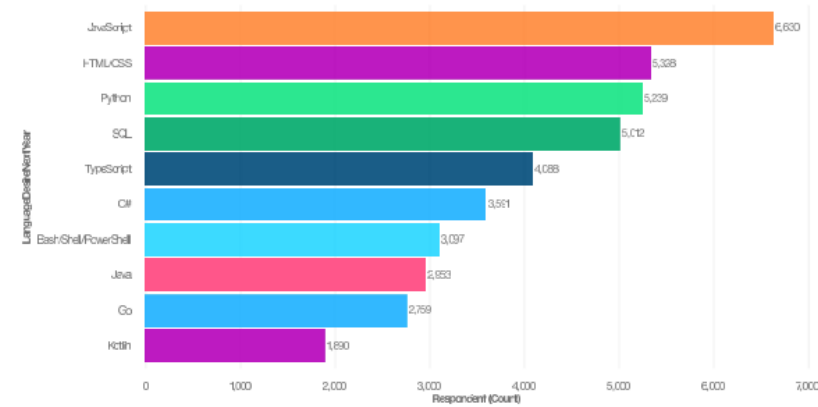


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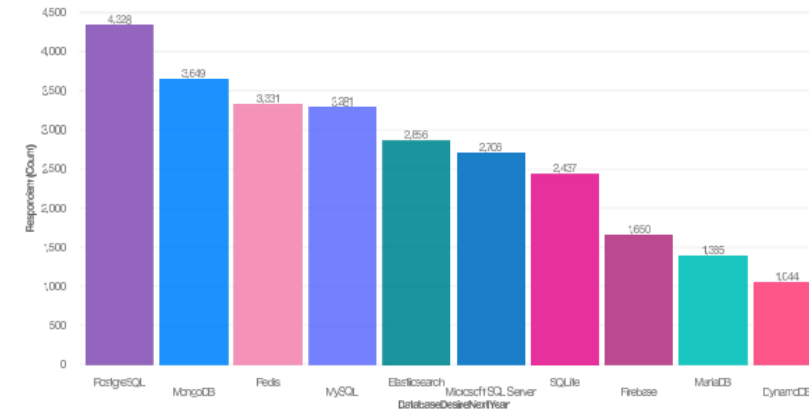


DASHBOARD TAB 2

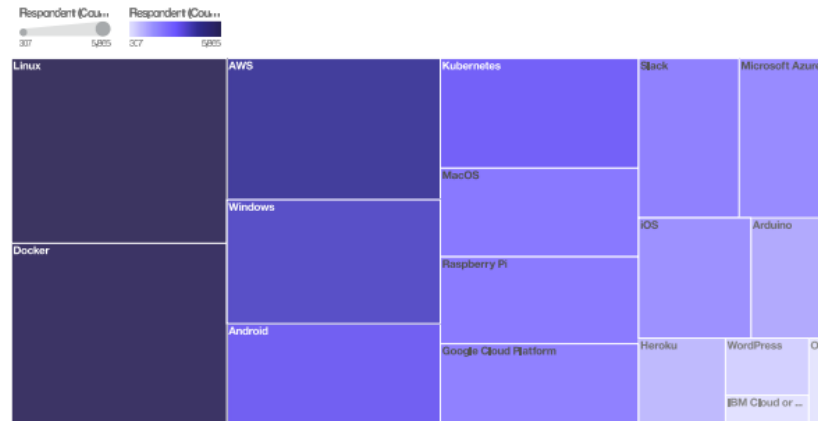
Top 10 Languages Desired to Work With Next Year



Top 10 Databases Desired to Work With Next Year



Platforms Desired to Work With Next Year



Top 10 Webframes Desired to Work With Next Year



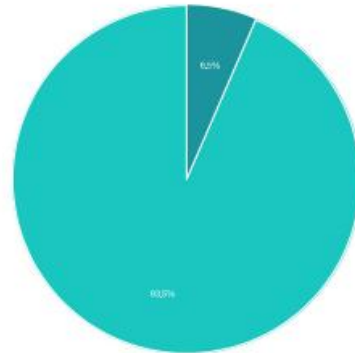
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DASHBOARD TAB 3

Percentage of Respondents based on Gender

Gender
Woman Man

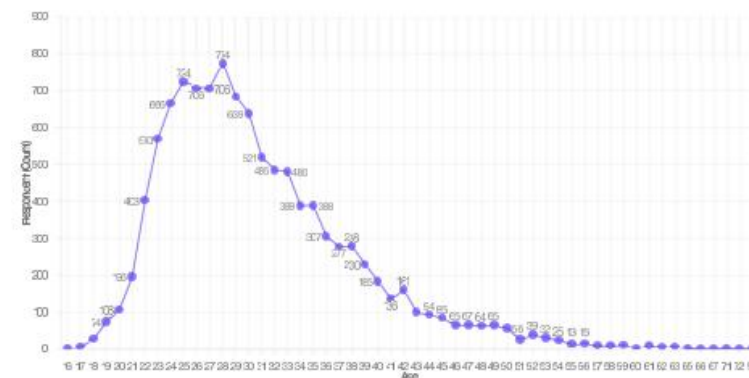


Respondents based on Location

Respondent (Count)
1 308

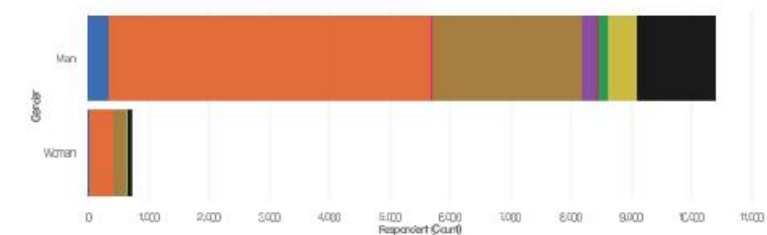


Respondents based on Age



Respondents by Gender and Education Level

Education Level
Associate degree Bachelor's degree (BA, BS, BE, ...) Incomplete certificate (GCE, ...) Master's degree (M.A., M.B.A., ...)
Other doctoral degree (Ph.D., ...) Primary/elementary school Professional degree (D.M.D., ...) Secondary school (e.g., ...)
Some college/University study ...



DISCUSSION

- Technology trends (Languages, Databases and Platforms).
- Demographic information of developers.



OVERALL FINDINGS & IMPLICATIONS

Findings

- There is a gender disparity for developers with only 6.5% as women and the rest 93.5% as men.
- The developers are concentrated in two countries: USA & India.
- Web Development Languages and Data Science & AI/ML languages are increasing popularity.

Implications

- Get the intention of invest to get more women attracted for being developers.
- Companies that are not located in USA and India, should consider outsourcing developers or investing in training local talent.
- Increment the investment for Data Science and AI/ML languages.



CONCLUSION



- Focus the investment and increment it for languages for Data Science and Machine Learning & AI.
- Attract more women for being developers.
- Invest in local developers or outsource devs located outside USA and India.

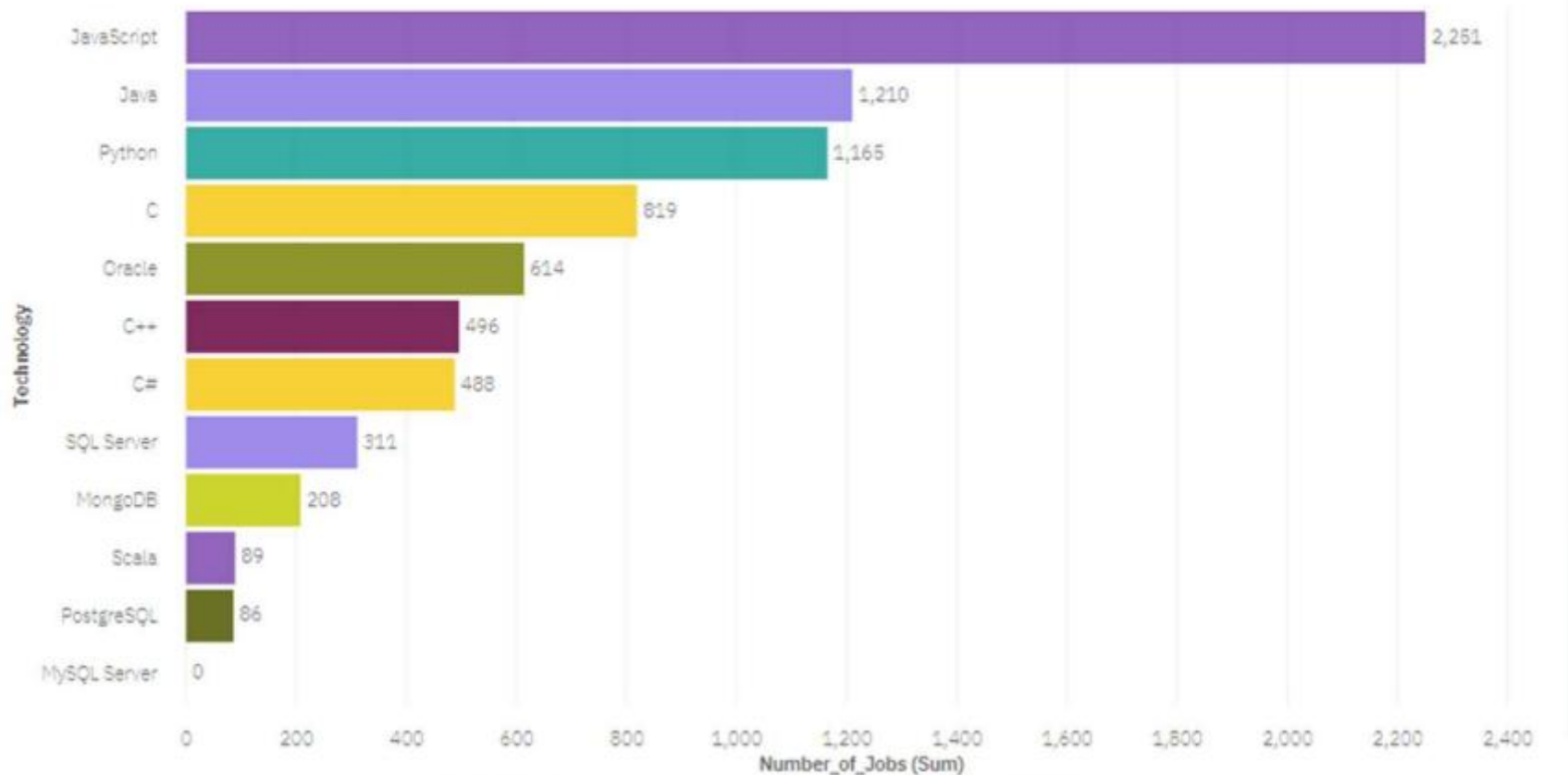


APPENDIX



JOB POSTING

Number of Job Postings for each Technology



POPULAR LANGUAGES

Average Annual Salary based on Language

